

COURSE SYLLABUS

< CSC15104> – < Data Recovery and Safety>

1. GENERAL INFORMATION

Course name:	Data Recovery and Safety
Course name (in Vietnamese):	An toàn và phục hồi dữ liệu
Course ID:	CSC15104
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Computer Systems, Operating Systems
Prior course:	Data structure

2. COURSE DESCRIPTION

Data or information plays important roles, eventually more than physical property; it does not exist then neither do lives. One of the most dangerous cases which would likely happen nowadays is data loss, whose consequences negatively affect any person, business or organization. This course provides basics of data organization of digital devices and introduces methods of data security , privacy and protection, as well as how to recover lost data.

3. COURSE GOALS

At the end of the course, students can

ID	Description	Program LOs
G1	Be able to understand the specialized English term and materials related to data recovery and safety	2.4.3, 2.4.5
G2	Practice presentation and report writing skills	2.2, 2.3.1, 2.3.2, 3.3

G3	Practice skills of discovering and developing methods to solve problems	2.2, 3.1, 3.3, 4.1, 4.2, 4.3
G4	Understand storage architecture and data organization mechanisms of storage devices, basic data formats. Discover and conduct experiments on methods of accessing data directly on devices.	1.1.1, 1.1.2, 1.1.3, 1.2, 1.3.1, 1.3.3, 1.3.6, 3.1, 5.1.1, 5.1.2, 5.2
G5	Understand and apply data recovery methods	1.2.1, 1.3.1, 1.3.3, 1.3.6, 2.2, 4.1, 5.2, 5.3, 6.1
G6	Understand and apply methods of protecting and securing data	1.2.1, 1.3.1, 1.3.3, 1.3.6
G7	Understand and organize a storage architecture given requirement of data safety	1.1.1, 1.2.1, 1.3.1

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Be able to understand specialized English terminology about data recovery and safety	T
G1.2	Understand English materials related to lectures.	U
G2.1	Practice writing and presentation skills at the request of the lecturer.	U
G2.2	Explain, demonstrate, defend ideas in the report	U
G3.1	Develop programs in C++ and data accessing tools	U
G3.2	Practice problem solving skills	I,U
G4.1	Apply math background for data organization	U
G4.2	Apply physical and electrical background for storage devices	I

G4.3	Practice reviewing, modifying, fixing, connecting storage devices	T,U
G4.4	Practice data discovery on common storage devices	T,U
G5.1	Understand mechanisms of storing and accessing data on different operating systems	T
G5.2	Apply data accessing tools to discover data, practice basic operations on data and recover data in simple formats.	T,U
G5.3	Apply understanding of data formats to develop programs of data recovery	T,U
G6.1	Understand circumstances affecting data safety	U
G6.2	Practice hardware engineering skills of data safety	T,U
G6.3	Practice software engineering skills of data safety	T,U
G7.1	Understand architecture of file systems	T
G7.2	Develop and organize file systems with basic operations	T,U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Overview of data safety and recovery	G1, G2, G3	Lecturing, Q&A, discussion	QZ1: Quiz 1
2	Data encryption, volume and device lock	G4.1, G4.2, G5.1, G6.3	Lecturing, demonstration, Q&A	HW1, PJ1

3	Common issues on storage devices and how to fix them	G1, G2, G4.1, G4.2, G6.2, G6.3	Lecturing, demonstration, discussion	QZ2: Quiz 2
4	Common file systems and architectures	G1, G4.1, G5, G6.1, G6.2	Lecturing, demonstration, discussion	HW2
5	Logical issues on file systems and how to fix them	G1, G2, G3, G4.4, G5.1, G5.2, G5.3	Lecturing, demonstration, case study & discussion	HW3 QZ3: Quiz 3
6	Data safety and recovery tools	G1, G2, G3, G5.1	Lecturing, demonstration, Q&A	Midterm test
7	Data recovery by formats	G1, G2, G3, G5.1	Lecturing, Demonstration, Q&A	PJ2
8	CDR techniques to remove abnormal content and recover data	G1, G2, G3, G5.1	Lecturing, Demonstration,	HW4
9	Develop mechanisms of data protecting	G1, G2, G3, G5.1, G6.1, G6.2, G6.3	Lecturing, Demonstration, Discussion	QZ4: Quiz 4
10	Build specific and independent file systems	G1, G2, G3, G5.1, G6.1, G7.1, G7.2	Lecturing, Demonstration,	PJ3
11	Review		Lecturing Q&A, Discussion	

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Data Encryption	G1, G2, G3	Explain, demonstrate, discuss	LW1, LW10
2	Volume and device lock	G1, G2, G3	Explain, demonstrate, discuss	LW2

3	Discover, practice skills of recovering data on common devices	G1, G2, G4.1, G4.2, G4.3	Explain, demonstrate,	LW3
4	Practice data safety tools	G1, G2, G3, G4, G5, G7.1	Explain, demonstrate,	LW4
5	Practice data recovery by formats	G1, G2, G3, G5.1	Explain, demonstrate,	LW5
6	Remove abnormal data by CDR	G1, G2, G3, G5.3	Explain, demonstrate,	LW6
7	Develop and build hardware encryption and software encryption to systems	G1, G2, G3, G5.1, G6.1, G7	Explain, demonstrate,	LW7
8	Build specific file systems	G1, G2, G3, G5.1, G6.1, G7	Explain, demonstrate,	LW8
9	Distribute data to multiple devices	G1, G2, G3, G5.1, G6.1, G7	Explain, demonstrate,	LW9
10	Seminar, project summary			

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
A11	Quizzes: QZ1, QZ2, QZ3, and QZ4.	Small quizzes in class for each topic		5%
A12	Homework: HW1, HW2, and HW3	HW2, HW3: reading comprehension and writing reports in English HW1, HW4: practicing based on knowledge taught in class		5%

A13	Weekly lab work: LW1-LW10			10%
A2	Projects			30%
A21	Project PJ1	Project for teamwork		10%
A21	Project PJ2	Project for teamwork		10%
A21	Project PJ3	Project for teamwork		10%
A3	Exams			50%
A31	Midterm exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems		15%
A32	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems		35%

7. RESOURCES

Textbooks

Data Recovery eBook - <https://www.easeus.com/download/data-recovery-book.pdf>

Safety & Security Handbook –

<https://reliefweb.int/sites/reliefweb.int/files/resources/care-safety-and-security-handbook.pdf>

Others

1. MSDN, <http://msdn.microsoft.com>
2. NTFS, FAT, <http://ntfs.com/>
3. Undelete a file in NTFS, <http://www.codeproject.com/file/NTFSUndelete.asp>
4. File Format, <http://www.wotsit.org>
5. Advanced System Format, <http://penta.zaniah.net/index.php?asf>
6. Linux-NTFS, <http://www.linux-ntfs.org/content/view/104/43/>

7. JPEG Header Format, http://www.fastgraph.com/help/jpeg_header_format.html
8. WAVE PCM sound file format,
<http://ccrma.stanford.edu/courses/422/projects/WaveFormat/>
9. ID3 tag, <http://id3.org>
10. Zip format, www.pkware.com/index.php?option=com_content&task=view&id=64&Itemid=107

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.